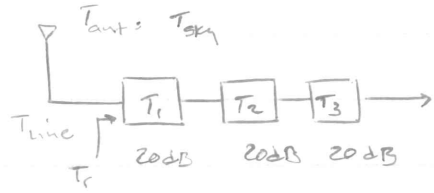
SAS 487
HOMEWORK 5
SOLUTIONS

1.



a) SDM = 164' $T_r = T_1 + \frac{T_2}{G_1} + \frac{T_3}{G_1 G_2} = 42.02$ K

$T_{sky} = 50$ $T_{line} = 300$

$T_{sys} = T_{sky} + \frac{1-\epsilon}{\epsilon} T_{line} + \frac{1}{\epsilon} T_r$ $\epsilon = e^{-\alpha_L L} = \frac{-.5\,dB}{100'} \times 164' = -8.2$ dB

$= 0.15$

$= 50 + 5.67 \times 300 + 6.67 \times 42 \sim \underline{\cancel{280}\ 2031\ K}$

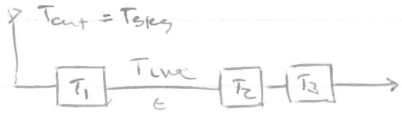
$\gg T_{sky}$ (NOT VERY GOOD)

b) TRY USING 26-17: $\epsilon = e^{-\alpha_L L} = \frac{-2.3\,dB}{100'} \cdot 164' = -3.77$ dB $= 0.42$

$T_{SYS} = 50 + 1.38 \times 300 + 2.38 \times 42 = \underline{564\ K}$

BETTER, BUT STILL $\gg T_{sky}$

c) NOW, TRY



RECALCULATE: $\left(\frac{S}{N}\right)_{OUT} = \frac{S_{in} G_1 G_2 G_3 \epsilon}{k_B B \left[T_{sky} G_1 \epsilon G_2 G_3 + T_1 \epsilon G_1 G_2 G_3 + (1-\epsilon) T_{line} G_2 G_3 + T_2 G_2 G_3 + T_3 G_3 \right]}$

$= \frac{S_{in}}{k_B B \left[T_{sky} + T_1 + \frac{1-\epsilon}{\epsilon} \frac{T_{line}}{G_1} + \frac{T_2}{\epsilon G_1} + \frac{T_3}{\epsilon G_1 G_2} \right]}$

SO $T_{sys} = T_{sky} + \underline{T_1} + \underline{\frac{1-\epsilon}{\epsilon} \frac{T_{line}}{G_1}} + \frac{T_2}{\epsilon G_1} + \frac{T_3}{\epsilon G_1 G_2}$

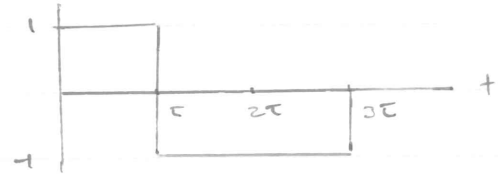
FRONT END REDUCED BY G_1

$= 50 + 40 + 5.67 \times \frac{300}{100} + 6.67 \times \left(\frac{200}{100} + \frac{200}{10{,}000} \right]$

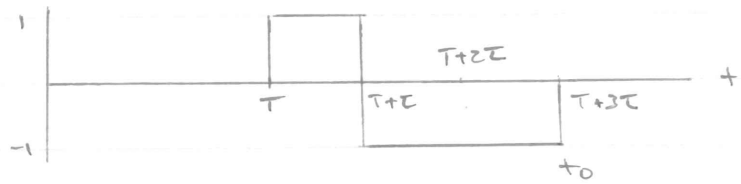
$\approx \underline{120\ K}$

MUCH BETTER STILL

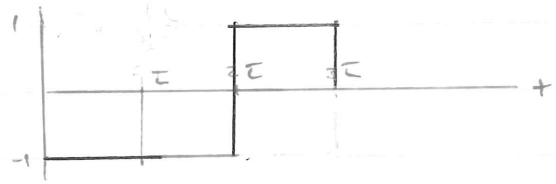
2 a) TRANSMIT



RECEIVE (FROM TARGET AT RANGE R, $T \equiv \frac{2R}{c}$)



b) $h(t) = f(t_0 - t)$ $t_0 = T + 3\tau$ LAST MOMENT SCATTER RECEIVED

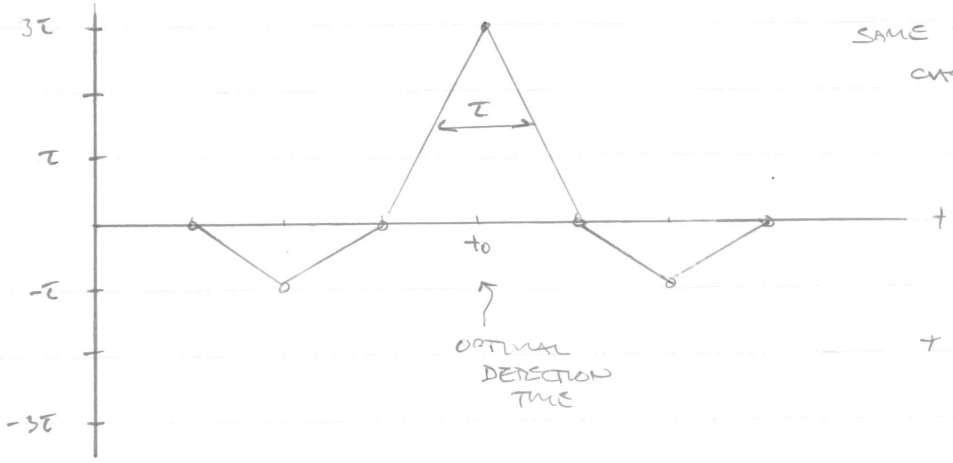


MATCHED FILTER IMPULSE RESPONSE FUNCTION

c) $g(t) = \int_{-\infty}^{t} f(t')\, h(t-t')\, dt'$ FILTER OUTPUT IS CONVOLUTION

$= \int_{-\infty}^{t} f(t')\, f(t' + t_0 - t)\, dt'$ ie AUTOCORRELATION OF f

AUTOCORRELATION MOST EASILY CALCULATED GRAPHICALLY



SAME FUNCTION AS CLASSROOM EXAMPLE

+ NOISE OUTPUT

PREPARED BY

DATE

3 RADAR $P_r = P_t \frac{D}{4\pi r^2} \sigma \frac{A_{eff}}{4\pi r^2}$ $D = \frac{4\pi}{\lambda^2} A_{eff}$

$= P_t \frac{A_{eff}^2}{4\pi r^2} \frac{\sigma}{\lambda^2}$ $A_{eff} = \pi 0.1^2 \cdot 0.6 = 1.88 \times 10^{-2} M^2$

$\lambda = 2.84$ CM

$P_N = k_B B T = 1.38 \times 10^{-17}$ W

FOR SNR = 10, $r^4 = \frac{P_t}{10} \frac{A_{eff}^2}{4\pi} \frac{\sigma}{\lambda^2} \frac{1}{P_N} = 5.05 \times 10^{12}$

$r \simeq 1.5$ km SEEMS REASONABLE

TX → ↓ RX TX ↗ ↓ RX

RADIO $P_r = P_t \frac{D_t}{4\pi r^2} A_r = P_t \frac{A_t}{4\pi r^2} D_r$

$P_N = k_B B T = 6.9 \times 10^{-16}$ W NOW

↓ 6 dB

SO $r^2 = \frac{P_t}{10} \frac{A_{eff,t} D_r}{4\pi} \frac{1}{P_N}$ $A_{eff,t}$ FROM ABOVE

$= 8.63 \times 10^{10}$

OR $r \simeq 294$ km!

ADVANTAGE IN RADIO CASE: $P_r \sim r^{-2}$, NOT r^{-4}

ALSO, NO σ INVOLVED

IN PRACTICE, NEED DIRECT LINE OF SIGHT IN RADIO CASE.

CURVATURE OF EARTH WOULD PREVENT 100 km RAY PATH.

EVEN MORE, ATMOSPHERIC ABSORPTION WOULD SEVERELY

LIMIT COMMUNICATION PATH LENGTH.

4 from class, know radiation pattern will be (uniform illumination)

$$F(\theta) = \frac{2 J_1(ka \sin\theta)}{ka \sin\theta}$$ normalized to unity

$$ka = \frac{2\pi}{\lambda} 25\lambda = 50\pi$$

$|F(\theta)|^2$ gives radiation pattern

$$HPBW = 1.02 \frac{\lambda}{2a} = \frac{1.02}{50} \text{ rad} = 1.17^\circ$$

$$A_{eff} = \pi (25\lambda)^2 = A_{phys} \qquad \varepsilon_{ap} = 1$$ } less if you consider actual reflector size

$$D_{max} = \frac{4\pi}{\lambda^2} A_{eff} = \frac{4\pi}{\lambda^2} \pi (25\lambda)^2 = 43.7 \text{ dB} = 2.47 \times 10^4$$

$= G_{max}$

$D(\theta) = |F|^2 D_{max}$ gives directivity vs. bearing

sidelobes down 17.6 dB

5) Backscatter gain $$G_{BS} = \frac{1}{4\pi} \int_0^{\pi/2} d\theta \int_0^{2\pi} d\phi \; G^2(\theta) \sin\theta$$

$$= \frac{1}{2} \int_0^{\pi/2} d\theta \; G^2(\theta) \sin\theta$$

$= 11365 = 40.6$ dB about half G_{max}

$$A_{BS} = \frac{\lambda^2}{4\pi} G_{BS} \cong \pi (17\lambda)^2 \equiv$$ ALSO ABOUT HALF OF $A_{PHYS} = A_{eff}$

ACTUAL PHYSICAL AREA IS $\pi \left(\frac{75}{2}\lambda\right)^2 > A_{eff} > A_{BS}$

$\sim 2 A_{BS}$